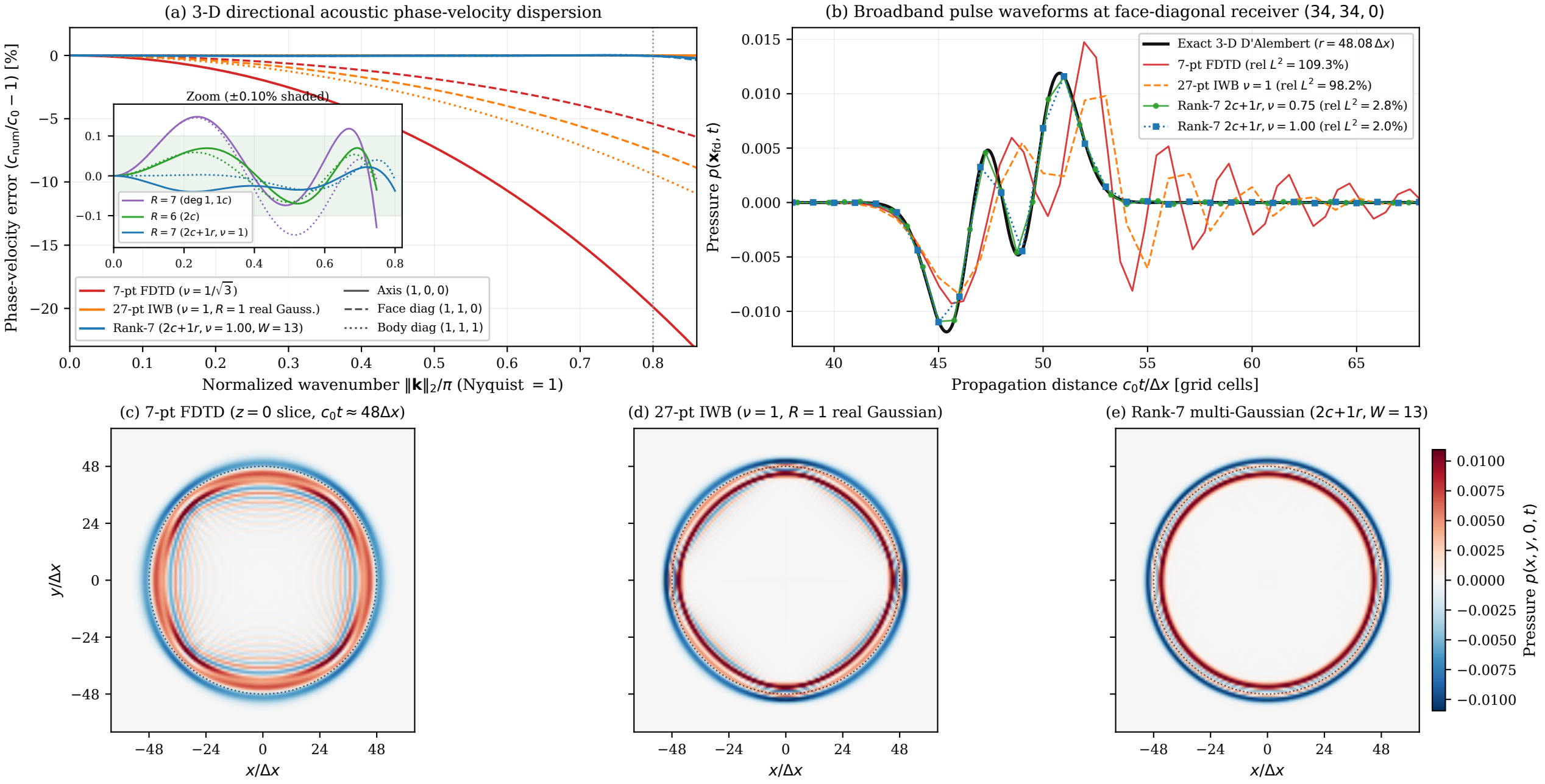

(a) 3-D directional acoustic phase-velocity dispersion
Phase-velocity error ($c_{num}/c_0 - 1$) [%]
Normalized wavenumber $\|\mathbf{k}\|_2/\pi$ (Nyquist = 1)
Zoom (±0.10% shaded)
R = 7 (deg 1, 1c)
R = 6 (2c)
R = 7 (2c+1r, ν = 1)
7-pt FDTD (ν = 1/√3)
27-pt IWB (ν = 1, R = 1 real Gauss.)
Rank-7 (2c+1r, ν = 1.00, W = 13)
Axis (1, 0, 0)
Face diag (1, 1, 0)
Body diag (1, 1, 1)
(b) Broadband pulse waveforms at face-diagonal receiver (34, 34, 0)
Pressure $p(\mathbf{x}_{fd}, t)$
Propagation distance $c_0 t/\Delta x$ [grid cells]
Exact 3-D D'Alembert ($r = 48.08\,\Delta x$)
7-pt FDTD (rel $L^2 = 109.3\%$)
27-pt IWB ν = 1 (rel $L^2 = 98.2\%$)
Rank-7 2c+1r, ν = 0.75 (rel $L^2 = 2.8\%$)
Rank-7 2c+1r, ν = 1.00 (rel $L^2 = 2.0\%$)
(c) 7-pt FDTD ($z = 0$ slice, $c_0 t \approx 48\Delta x$)
(d) 27-pt IWB ($\nu = 1$, $R = 1$ real Gaussian)
(e) Rank-7 multi-Gaussian ($2c+1r, W = 13$)
$y/\Delta x$
$x/\Delta x$
Pressure $p(x, y, 0, t)$